

BIDAR UNIVERSITY BIDAR

ZOOLOGY V SEMESTER

DSC-5 (THEORY): Developmental Biology, Genetics and Evolution

Course Duration: 60 Hours

Credits: 04

Unit I: Developmental Biology

22 Hours

- **Gametogenesis:** Spermatogenesis and Oogenesis in mammals.
- **Eggs:** Types of eggs based on quantity and distribution of yolk with examples.
- **Fertilization:** External (Amphibians) & Internal (Mammals), Monospermy & Polyspermy, Significance of fertilization.
- **Early Development:** Cleavage, Morula, Blastula, Gastrula & Neurula.
- **Morphogenetic Movements & Germ Layers:** Fate of germ layers (Frog embryo).
- **Development of Chick:** Structure of hen's egg, origin & development of primitive streak (18 hrs & 24 hrs).
- **Extra-Embryonic Membranes:** Structure & functions of amnion, chorion, yolk sac, and allantois.
- **Human Development:** Structure of mature spermatozoan, Menstrual cycle, Graafian follicle, ovulation, fertilization, blastocyst, implantation.
- **Placenta:** Histological and morphological classification with examples; Placental hormones.
- **Organogenesis:** Outline derivatives of different germ layers.

Unit II: Genetics

22 Hours

- **Introduction:** Definition, history of genetics, branches of genetics, heredity and variation.
- **Mendelian Genetics and Its Extension:** Principles of inheritance (Monohybrid, Dihybrid, and Laws of inheritance), Incomplete dominance, and Co-dominance.
- **Interaction of Genes:**
 - **Supplementary factors (9:3:3:1)** e.g., Comb pattern in fowls.
 - **Complementary factors (9:7)** - e.g., Flower colour in sweet peas.
 - **Dominant Epistasis (12:3:1)** - e.g., Plumage colour in Leghorn and Wyandotte.
 - **Recessive Epistasis (9:3:4)** - e.g., Coat colour in pigs.
- **Multiple Alleles:** ABO blood groups in human, Rh-factor, and Lethal alleles.
- **Sex Determination:** Chromosomal mechanism of sex determination, Genic balance theory, Klinefelter's and Turner's syndromes, Sex-linked inheritance in man (Haemophilia, Colour blindness), Gynandromorphs & Intersex, and Y-linked genes.
- **Mutations:** Chromosomal mutations and Gene mutation.
- **Linkage & Crossing-Over:** Theories of linkage and crossing over.
- **Human Genetics:** Common human genetic disorders in man, Inborn errors of metabolism (Albinism, Phenylketonuria, Alkaptonuria, Sickle cell anemia, Thalassemia), and Eugenics.

Unit III: Evolution

16 Hours

- **Evolutionary Theories:** Lamarckism, Darwinism, and Neo-Darwinism.
- **Species Concept & Organic Evolution:** Concept of organic evolution.
- **Evidences for Organic Evolution:**
 - **Indirect Evidences:** Comparative morphology, anatomy, connecting links (Peripatus, Neoceratodus, Archaeopteryx), Homology, Analogy, Vestigial organs (Human).
 - **Direct Evidences:** Paleontology - Nature & types of fossils; Brief account of Dinosaurs.
- **Macro-evolutionary Principles:** Adaptive radiation (e.g., Darwin's Finches).
- Hardy-Weinberg Law.
- Evolutionary History of Man.

B.SC. SEMESTER V EXAMINATION

Zoology | Theory Question Paper Structure & Practical Syllabus

Paper Code:	ZT-5	Subject:	Developmental Biology, Genetics & Evolution
Course:	B.Sc. Semester V	Practical Code:	DSC-2P-5

1. THEORY EXAM MODEL QUESTION PAPER STRUCTURE

Duration:	3 Hours	Max Marks:	80 Marks
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Instructions to Candidates:

- Answer all sections as per specific section requirements.
- Draw neat, labeled diagrams wherever necessary.

Section	Question Format / Type	Questions to Answer	Marks Distribution
Section - A	Short answer questions (Questions 1 to 12)	Answer Any 10 out of 12	10 × 2 Marks = 20 Marks
Section - B	Medium long answer questions (Questions 13 to 18)	Answer Any 4 out of 6	4 × 5 Marks = 20 Marks
Section - C	Detailed essay-type questions (Questions 19 to 24)	Answer Any 4 out of 6	4 × 10 Marks = 40 Marks
Total Marks			80 Marks

2. PRACTICAL SYLLABUS (DSC-2P-5: 02 CREDITS)

Course Title: Practical Developmental Biology, Genetics and Evolution

- Blood Group Analysis:** Detection of Blood Group (A, B, AB, O) and Rh-factor. Write down their genetic significance.
- Genetics Problems:** Monohybrid cross, Dihybrid cross, Sex-linked inheritance, Multiple alleles, and Pedigree analysis.
- Study of Homologous Organs:** Comparative study of Forelimbs of Frog and Bird.
- Study of Analogous Organs:** Comparative study of Wings of Insects and Vertebrate wings (Bird/Bat).
- Study of Vestigial Organs:** Study of Vermiform Appendix and Coccyx in humans.
- Study of Paleontological Models:**
 - Models of Dinosaurs: *Ichthyosaurus*, *Brontosaurus*, and *Stegosaurus*.
 - Fossil Bird model: *Archaeopteryx*.
 - Fossil Man models (Any four available models).

7. **Developmental Stages of Frog:** Study of Cleavage stages, Blastula, Gastrula, and various stages of Tadpole larvae.
8. **Permanent Slides of Chick Embryo:** Microscopic study of 18 Hours, 24 Hours, 36 Hours, and 48 Hours chick embryos.
9. **Chick Embryo Mounting:** Preparation of whole mount of chick embryo.

3. PRACTICAL EXAMINATION QUESTION PAPER FORMAT

Practical Exam Duration: 3 Hours | Total Marks: 40 Marks

Q. No.	Practical Task / Question Topic	Marks Allocated
1	Mounting of Chick Embryo	07 Marks
2	Identification of Embryological Slides (2 Frog slides + 1 Chick slide)	02 × 3 = 06 Marks
3	Detection of Blood Group (A, B, AB, O) & Rh-factor with Genetic Significance	05 Marks
4	Genetics Problem solving (Monohybrid / Dihybrid / Sex-linked inheritance / Multiple alleles)	05 Marks
5	Identification - Evolutionary Organs (Homologous / Analogous / Vestigial organ)	03 × 2 = 06 Marks
6	Identification - Paleontology (Model of Dinosaur / <i>Archaeopteryx</i> / Fossil Man)	03 × 2 = 06 Marks
7	Practical Record Book & viva-voce (Certified Record)	05 Marks
Total Practical Marks:		40 Marks

4. SUGGESTED REFERENCE BOOKS

- **Developmental Biology** — Gilbert, S. F. (2008)
- **Patten's Foundations of Embryology** — Carlson, Bruce M. (2008)
- **An Introduction to Embryology** — Balinsky, B. I.
- **Principles of Genetics** — Gardner, E. J., Simmons, M. J., & Snustad, D. P. (8th Edition, 2008)
- **Concepts of Genetics** — Klug, W. S., Cummings, M. R., & Spencer, C. A. (2012)
- **Evolutionary Biology** — Futuyma, Douglas J. (1997)
- **Evolution** — Hall, B. K. & Hallgrímsson, B. (2008)